

B.5-2

Let $G = (V, E)$ be a directed acyclic graph in which there is a vertex $v_0 \in V$ such that there exists a unique path from v_0 to every vertex $v \in V$. Prove that the undirected version of G forms a tree.

B.5-3

Show by induction that the number of degree-2 nodes in any nonempty binary tree is 1 fewer than the number of leaves. Conclude that the number of internal nodes in a full binary tree is 1 fewer than the number of leaves.

B.5-4

Use induction to show that a nonempty binary tree with n nodes has height at least $\lfloor \lg n \rfloor$.

B.5-5 ★

The **internal path length** of a full binary tree is the sum, taken over all internal nodes of the tree, of the depth of each node. Likewise, the **external path length** is the sum, taken over all leaves of the tree, of the depth of each leaf. Consider a full binary tree with n internal nodes, internal path length i , and external path length e . Prove that $e = i + 2n$.

B.5-6 ★

Let us associate a “weight” $w(x) = 2^{-d}$ with each leaf x of depth d in a binary tree T , and let L be the set of leaves of T . Prove that $\sum_{x \in L} w(x) \leq 1$. (This is known as the **Kraft inequality**.)

B.5-7 ★

Show that if $L \geq 2$, then every binary tree with L leaves contains a subtree having between $L/3$ and $2L/3$ leaves, inclusive.

Problems
B-1 Graph coloring

Given an undirected graph $G = (V, E)$, a **k -coloring** of G is a function $c : V \rightarrow \{0, 1, \dots, k-1\}$ such that $c(u) \neq c(v)$ for every edge $(u, v) \in E$. In other words, the numbers $0, 1, \dots, k-1$ represent the k colors, and adjacent vertices must have different colors.

a. Show that any tree is 2-colorable.

- b.** Show that the following are equivalent:
1. G is bipartite.
 2. G is 2-colorable.
 3. G has no cycles of odd length.
- c.** Let d be the maximum degree of any vertex in a graph G . Prove that we can color G with $d + 1$ colors.
- d.** Show that if G has $O(|V|)$ edges, then we can color G with $O(\sqrt{|V|})$ colors.

B-2 Friendly graphs

Reword each of the following statements as a theorem about undirected graphs, and then prove it. Assume that friendship is symmetric but not reflexive.

- a.** Any group of at least two people contains at least two people with the same number of friends in the group.
- b.** Every group of six people contains either at least three mutual friends or at least three mutual strangers.
- c.** Any group of people can be partitioned into two subgroups such that at least half the friends of each person belong to the subgroup of which that person is *not* a member.
- d.** If everyone in a group is the friend of at least half the people in the group, then the group can be seated around a table in such a way that everyone is seated between two friends.

B-3 Bisecting trees

Many divide-and-conquer algorithms that operate on graphs require that the graph be bisected into two nearly equal-sized subgraphs, which are induced by a partition of the vertices. This problem investigates bisections of trees formed by removing a small number of edges. We require that whenever two vertices end up in the same subtree after removing edges, then they must be in the same partition.

- a.** Show that we can partition the vertices of any n -vertex binary tree into two sets A and B , such that $|A| \leq 3n/4$ and $|B| \leq 3n/4$, by removing a single edge.
- b.** Show that the constant $3/4$ in part (a) is optimal in the worst case by giving an example of a simple binary tree whose most evenly balanced partition upon removal of a single edge has $|A| = 3n/4$.

- c. Show that by removing at most $O(\lg n)$ edges, we can partition the vertices of any n -vertex binary tree into two sets A and B such that $|A| = \lfloor n/2 \rfloor$ and $|B| = \lceil n/2 \rceil$.

Appendix notes

G. Boole pioneered the development of symbolic logic, and he introduced many of the basic set notations in a book published in 1854. Modern set theory was created by G. Cantor during the period 1874–1895. Cantor focused primarily on sets of infinite cardinality. The term “function” is attributed to G. W. Leibniz, who used it to refer to several kinds of mathematical formulas. His limited definition has been generalized many times. Graph theory originated in 1736, when L. Euler proved that it was impossible to cross each of the seven bridges in the city of Königsberg exactly once and return to the starting point.

The book by Harary [160] provides a useful compendium of many definitions and results from graph theory.

C Counting and Probability

This appendix reviews elementary combinatorics and probability theory. If you have a good background in these areas, you may want to skim the beginning of this appendix lightly and concentrate on the later sections. Most of this book's chapters do not require probability, but for some chapters it is essential.

Section C.1 reviews elementary results in counting theory, including standard formulas for counting permutations and combinations. The axioms of probability and basic facts concerning probability distributions form Section C.2. Random variables are introduced in Section C.3, along with the properties of expectation and variance. Section C.4 investigates the geometric and binomial distributions that arise from studying Bernoulli trials. The study of the binomial distribution continues in Section C.5, an advanced discussion of the “tails” of the distribution.

C.1 Counting

Counting theory tries to answer the question “How many?” without actually enumerating all the choices. For example, we might ask, “How many different n -bit numbers are there?” or “How many orderings of n distinct elements are there?” In this section, we review the elements of counting theory. Since some of the material assumes a basic understanding of sets, you might wish to start by reviewing the material in Section B.1.

Rules of sum and product

We can sometimes express a set of items that we wish to count as a union of disjoint sets or as a Cartesian product of sets.

The **rule of sum** says that the number of ways to choose one element from one of two *disjoint* sets is the sum of the cardinalities of the sets. That is, if A and B are two finite sets with no members in common, then $|A \cup B| = |A| + |B|$, which

follows from equation (B.3). For example, each position on a car's license plate is a letter or a digit. The number of possibilities for each position is therefore $26 + 10 = 36$, since there are 26 choices if it is a letter and 10 choices if it is a digit.

The **rule of product** says that the number of ways to choose an ordered pair is the number of ways to choose the first element times the number of ways to choose the second element. That is, if A and B are two finite sets, then $|A \times B| = |A| \cdot |B|$, which is simply equation (B.4). For example, if an ice-cream parlor offers 28 flavors of ice cream and 4 toppings, the number of possible sundaes with one scoop of ice cream and one topping is $28 \cdot 4 = 112$.

Strings

A **string** over a finite set S is a sequence of elements of S . For example, there are 8 binary strings of length 3:

000, 001, 010, 011, 100, 101, 110, 111 .

We sometimes call a string of length k a **k -string**. A **substring** s' of a string s is an ordered sequence of consecutive elements of s . A **k -substring** of a string is a substring of length k . For example, 010 is a 3-substring of 01101001 (the 3-substring that begins in position 4), but 111 is not a substring of 01101001.

We can view a k -string over a set S as an element of the Cartesian product S^k of k -tuples; thus, there are $|S|^k$ strings of length k . For example, the number of binary k -strings is 2^k . Intuitively, to construct a k -string over an n -set, we have n ways to pick the first element; for each of these choices, we have n ways to pick the second element; and so forth k times. This construction leads to the k -fold product $n \cdot n \cdots n = n^k$ as the number of k -strings.

Permutations

A **permutation** of a finite set S is an ordered sequence of all the elements of S , with each element appearing exactly once. For example, if $S = \{a, b, c\}$, then S has 6 permutations:

$abc, acb, bac, bca, cab, cba$.

There are $n!$ permutations of a set of n elements, since we can choose the first element of the sequence in n ways, the second in $n - 1$ ways, the third in $n - 2$ ways, and so on.

A **k -permutation** of S is an ordered sequence of k elements of S , with no element appearing more than once in the sequence. (Thus, an ordinary permutation is an n -permutation of an n -set.) The twelve 2-permutations of the set $\{a, b, c, d\}$ are

$ab, ac, ad, ba, bc, bd, ca, cb, cd, da, db, dc$.

The number of k -permutations of an n -set is

$$n(n-1)(n-2)\cdots(n-k+1) = \frac{n!}{(n-k)!} , \quad (\text{C.1})$$

since we have n ways to choose the first element, $n-1$ ways to choose the second element, and so on, until we have selected k elements, the last being a selection from the remaining $n-k+1$ elements.

Combinations

A **k -combination** of an n -set S is simply a k -subset of S . For example, the 4-set $\{a, b, c, d\}$ has six 2-combinations:

ab, ac, ad, bc, bd, cd .

(Here we use the shorthand of denoting the 2-subset $\{a, b\}$ by ab , and so on.) We can construct a k -combination of an n -set by choosing k distinct (different) elements from the n -set. The order in which we select the elements does not matter.

We can express the number of k -combinations of an n -set in terms of the number of k -permutations of an n -set. Every k -combination has exactly $k!$ permutations of its elements, each of which is a distinct k -permutation of the n -set. Thus, the number of k -combinations of an n -set is the number of k -permutations divided by $k!$; from equation (C.1), this quantity is

$$\frac{n!}{k!(n-k)!} . \quad (\text{C.2})$$

For $k=0$, this formula tells us that the number of ways to choose 0 elements from an n -set is 1 (not 0), since $0! = 1$.

Binomial coefficients

The notation $\binom{n}{k}$ (read “ n choose k ”) denotes the number of k -combinations of an n -set. From equation (C.2), we have

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} .$$

This formula is symmetric in k and $n-k$:

$$\binom{n}{k} = \binom{n}{n-k} . \quad (\text{C.3})$$

These numbers are also known as **binomial coefficients**, due to their appearance in the **binomial expansion**:

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}. \quad (\text{C.4})$$

A special case of the binomial expansion occurs when $x = y = 1$:

$$2^n = \sum_{k=0}^n \binom{n}{k}.$$

This formula corresponds to counting the 2^n binary n -strings by the number of 1s they contain: $\binom{n}{k}$ binary n -strings contain exactly k 1s, since we have $\binom{n}{k}$ ways to choose k out of the n positions in which to place the 1s.

Many identities involve binomial coefficients. The exercises at the end of this section give you the opportunity to prove a few.

Binomial bounds

We sometimes need to bound the size of a binomial coefficient. For $1 \leq k \leq n$, we have the lower bound

$$\begin{aligned} \binom{n}{k} &= \frac{n(n-1) \cdots (n-k+1)}{k(k-1) \cdots 1} \\ &= \left(\frac{n}{k}\right) \left(\frac{n-1}{k-1}\right) \cdots \left(\frac{n-k+1}{1}\right) \\ &\geq \left(\frac{n}{k}\right)^k. \end{aligned}$$

Taking advantage of the inequality $k! \geq (k/e)^k$ derived from Stirling's approximation (3.18), we obtain the upper bounds

$$\begin{aligned} \binom{n}{k} &= \frac{n(n-1) \cdots (n-k+1)}{k(k-1) \cdots 1} \\ &\leq \frac{n^k}{k!} \\ &\leq \left(\frac{en}{k}\right)^k. \end{aligned} \quad (\text{C.5})$$

For all integers k such that $0 \leq k \leq n$, we can use induction (see Exercise C.1-12) to prove the bound

$$\binom{n}{k} \leq \frac{n^n}{k^k (n-k)^{n-k}}, \quad (\text{C.6})$$

where for convenience we assume that $0^0 = 1$. For $k = \lambda n$, where $0 \leq \lambda \leq 1$, we can rewrite this bound as

$$\begin{aligned} \binom{n}{\lambda n} &\leq \frac{n^n}{(\lambda n)^{\lambda n} ((1-\lambda)n)^{(1-\lambda)n}} \\ &= \left(\left(\frac{1}{\lambda} \right)^\lambda \left(\frac{1}{1-\lambda} \right)^{1-\lambda} \right)^n \\ &= 2^{n H(\lambda)}, \end{aligned}$$

where

$$H(\lambda) = -\lambda \lg \lambda - (1-\lambda) \lg(1-\lambda) \quad (\text{C.7})$$

is the **(binary) entropy function** and where, for convenience, we assume that $0 \lg 0 = 0$, so that $H(0) = H(1) = 0$.

Exercises

C.1-1

How many k -substrings does an n -string have? (Consider identical k -substrings at different positions to be different.) How many substrings does an n -string have in total?

C.1-2

An n -input, m -output **boolean function** is a function from $\{\text{TRUE}, \text{FALSE}\}^n$ to $\{\text{TRUE}, \text{FALSE}\}^m$. How many n -input, 1-output boolean functions are there? How many n -input, m -output boolean functions are there?

C.1-3

In how many ways can n professors sit around a circular conference table? Consider two seatings to be the same if one can be rotated to form the other.

C.1-4

In how many ways can we choose three distinct numbers from the set $\{1, 2, \dots, 99\}$ so that their sum is even?

C.1-5

Prove the identity

$$\binom{n}{k} = \frac{n}{k} \binom{n-1}{k-1} \quad (\text{C.8})$$

for $0 < k \leq n$.

C.1-6

Prove the identity

$$\binom{n}{k} = \frac{n}{n-k} \binom{n-1}{k}$$

for $0 \leq k < n$.

C.1-7

To choose k objects from n , you can make one of the objects distinguished and consider whether the distinguished object is chosen. Use this approach to prove that

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

C.1-8

Using the result of Exercise C.1-7, make a table for $n = 0, 1, \dots, 6$ and $0 \leq k \leq n$ of the binomial coefficients $\binom{n}{k}$ with $\binom{0}{0}$ at the top, $\binom{1}{0}$ and $\binom{1}{1}$ on the next line, and so forth. Such a table of binomial coefficients is called **Pascal's triangle**.

C.1-9

Prove that

$$\sum_{i=1}^n i = \binom{n+1}{2}.$$

C.1-10

Show that for any integers $n \geq 0$ and $0 \leq k \leq n$, the expression $\binom{n}{k}$ achieves its maximum value when $k = \lfloor n/2 \rfloor$ or $k = \lceil n/2 \rceil$.

C.1-11 ★

Argue that for any integers $n \geq 0$, $j \geq 0$, $k \geq 0$, and $j + k \leq n$,

$$\binom{n}{j+k} \leq \binom{n}{j} \binom{n-j}{k}. \quad (\text{C.9})$$

Provide both an algebraic proof and an argument based on a method for choosing $j + k$ items out of n . Give an example in which equality does not hold.

C.1-12 ★

Use induction on all integers k such that $0 \leq k \leq n/2$ to prove inequality (C.6), and use equation (C.3) to extend it to all integers k such that $0 \leq k \leq n$.

C.1-13 ★

Use Stirling's approximation to prove that

$$\binom{2n}{n} = \frac{2^{2n}}{\sqrt{\pi n}} (1 + O(1/n)) . \quad (\text{C.10})$$

C.1-14 ★

By differentiating the entropy function $H(\lambda)$, show that it achieves its maximum value at $\lambda = 1/2$. What is $H(1/2)$?

C.1-15 ★

Show that for any integer $n \geq 0$,

$$\sum_{k=0}^n \binom{n}{k} k = n 2^{n-1} . \quad (\text{C.11})$$

C.2 Probability

Probability is an essential tool for the design and analysis of probabilistic and randomized algorithms. This section reviews basic probability theory.

We define probability in terms of a **sample space** S , which is a set whose elements are called **elementary events**. We can think of each elementary event as a possible outcome of an experiment. For the experiment of flipping two distinguishable coins, with each individual flip resulting in a head (H) or a tail (T), we can view the sample space as consisting of the set of all possible 2-strings over $\{H, T\}$:

$$S = \{HH, HT, TH, TT\} .$$

An **event** is a subset¹ of the sample space S . For example, in the experiment of flipping two coins, the event of obtaining one head and one tail is $\{HT, TH\}$. The event S is called the **certain event**, and the event $\emptyset$ is called the **null event**. We say that two events A and B are **mutually exclusive** if $A \cap B = \emptyset$. We sometimes treat an elementary event $s \in S$ as the event $\{s\}$. By definition, all elementary events are mutually exclusive.

Axioms of probability

A **probability distribution** $\Pr\{\cdot\}$ on a sample space S is a mapping from events of S to real numbers satisfying the following **probability axioms**:

1. $\Pr\{A\} \geq 0$ for any event A .
2. $\Pr\{S\} = 1$.
3. $\Pr\{A \cup B\} = \Pr\{A\} + \Pr\{B\}$ for any two mutually exclusive events A and B . More generally, for any (finite or countably infinite) sequence of events $A_1, A_2, \dots$ that are pairwise mutually exclusive,

$$\Pr\left\{\bigcup_i A_i\right\} = \sum_i \Pr\{A_i\}.$$

We call $\Pr\{A\}$ the **probability** of the event A . We note here that axiom 2 is a normalization requirement: there is really nothing fundamental about choosing 1 as the probability of the certain event, except that it is natural and convenient.

Several results follow immediately from these axioms and basic set theory (see Section B.1). The null event $\emptyset$ has probability $\Pr\{\emptyset\} = 0$. If $A \subseteq B$, then $\Pr\{A\} \leq \Pr\{B\}$. Using $\overline{A}$ to denote the event $S - A$ (the **complement** of A), we have $\Pr\{\overline{A}\} = 1 - \Pr\{A\}$. For any two events A and B ,

$$\Pr\{A \cup B\} = \Pr\{A\} + \Pr\{B\} - \Pr\{A \cap B\} \tag{C.12}$$

$$\leq \Pr\{A\} + \Pr\{B\}. \tag{C.13}$$

¹For a general probability distribution, there may be some subsets of the sample space S that are not considered to be events. This situation usually arises when the sample space is uncountably infinite. The main requirement for what subsets are events is that the set of events of a sample space be closed under the operations of taking the complement of an event, forming the union of a finite or countable number of events, and taking the intersection of a finite or countable number of events. Most of the probability distributions we shall see are over finite or countable sample spaces, and we shall generally consider all subsets of a sample space to be events. A notable exception is the continuous uniform probability distribution, which we shall see shortly.

In our coin-flipping example, suppose that each of the four elementary events has probability $1/4$. Then the probability of getting at least one head is

$$\begin{aligned}\Pr\{HH, HT, TH\} &= \Pr\{HH\} + \Pr\{HT\} + \Pr\{TH\} \\ &= 3/4.\end{aligned}$$

Alternatively, since the probability of getting strictly less than one head is $\Pr\{TT\} = 1/4$, the probability of getting at least one head is $1 - 1/4 = 3/4$.

Discrete probability distributions

A probability distribution is **discrete** if it is defined over a finite or countably infinite sample space. Let S be the sample space. Then for any event A ,

$$\Pr\{A\} = \sum_{s \in A} \Pr\{s\},$$

since elementary events, specifically those in A , are mutually exclusive. If S is finite and every elementary event $s \in S$ has probability

$$\Pr\{s\} = 1/|S|,$$

then we have the **uniform probability distribution** on S . In such a case the experiment is often described as “picking an element of S at random.”

As an example, consider the process of flipping a **fair coin**, one for which the probability of obtaining a head is the same as the probability of obtaining a tail, that is, $1/2$. If we flip the coin n times, we have the uniform probability distribution defined on the sample space $S = \{H, T\}^n$, a set of size 2^n . We can represent each elementary event in S as a string of length n over $\{H, T\}$, each string occurring with probability $1/2^n$. The event

$$A = \{\text{exactly } k \text{ heads and exactly } n - k \text{ tails occur}\}$$

is a subset of S of size $|A| = \binom{n}{k}$, since $\binom{n}{k}$ strings of length n over $\{H, T\}$ contain exactly k H's. The probability of event A is thus $\Pr\{A\} = \binom{n}{k}/2^n$.

Continuous uniform probability distribution

The continuous uniform probability distribution is an example of a probability distribution in which not all subsets of the sample space are considered to be events. The continuous uniform probability distribution is defined over a closed interval $[a, b]$ of the reals, where $a < b$. Our intuition is that each point in the interval $[a, b]$ should be “equally likely.” There are an uncountable number of points, however, so if we give all points the same finite, positive probability, we cannot simultaneously satisfy axioms 2 and 3. For this reason, we would like to associate a

probability only with *some* of the subsets of S , in such a way that the axioms are satisfied for these events.

For any closed interval $[c, d]$, where $a \leq c \leq d \leq b$, the **continuous uniform probability distribution** defines the probability of the event $[c, d]$ to be

$$\Pr\{[c, d]\} = \frac{d - c}{b - a}.$$

Note that for any point $x = [x, x]$, the probability of x is 0. If we remove the endpoints of an interval $[c, d]$, we obtain the open interval (c, d) . Since $[c, d] = [c, c] \cup (c, d) \cup [d, d]$, axiom 3 gives us $\Pr\{[c, d]\} = \Pr\{(c, d)\}$. Generally, the set of events for the continuous uniform probability distribution contains any subset of the sample space $[a, b]$ that can be obtained by a finite or countable union of open and closed intervals, as well as certain more complicated sets.

Conditional probability and independence

Sometimes we have some prior partial knowledge about the outcome of an experiment. For example, suppose that a friend has flipped two fair coins and has told you that at least one of the coins showed a head. What is the probability that both coins are heads? The information given eliminates the possibility of two tails. The three remaining elementary events are equally likely, so we infer that each occurs with probability $1/3$. Since only one of these elementary events shows two heads, the answer to our question is $1/3$.

Conditional probability formalizes the notion of having prior partial knowledge of the outcome of an experiment. The **conditional probability** of an event A given that another event B occurs is defined to be

$$\Pr\{A \mid B\} = \frac{\Pr\{A \cap B\}}{\Pr\{B\}} \quad (\text{C.14})$$

whenever $\Pr\{B\} \neq 0$. (We read “ $\Pr\{A \mid B\}$ ” as “the probability of A given B .”) Intuitively, since we are given that event B occurs, the event that A also occurs is $A \cap B$. That is, $A \cap B$ is the set of outcomes in which both A and B occur. Because the outcome is one of the elementary events in B , we normalize the probabilities of all the elementary events in B by dividing them by $\Pr\{B\}$, so that they sum to 1. The conditional probability of A given B is, therefore, the ratio of the probability of event $A \cap B$ to the probability of event B . In the example above, A is the event that both coins are heads, and B is the event that at least one coin is a head. Thus, $\Pr\{A \mid B\} = (1/4)/(3/4) = 1/3$.

Two events are **independent** if

$$\Pr\{A \cap B\} = \Pr\{A\} \Pr\{B\}, \quad (\text{C.15})$$

which is equivalent, if $\Pr\{B\} \neq 0$, to the condition

$$\Pr\{A \mid B\} = \Pr\{A\} .$$

For example, suppose that we flip two fair coins and that the outcomes are independent. Then the probability of two heads is $(1/2)(1/2) = 1/4$. Now suppose that one event is that the first coin comes up heads and the other event is that the coins come up differently. Each of these events occurs with probability $1/2$, and the probability that both events occur is $1/4$; thus, according to the definition of independence, the events are independent—even though you might think that both events depend on the first coin. Finally, suppose that the coins are welded together so that they both fall heads or both fall tails and that the two possibilities are equally likely. Then the probability that each coin comes up heads is $1/2$, but the probability that they both come up heads is $1/2 \neq (1/2)(1/2)$. Consequently, the event that one comes up heads and the event that the other comes up heads are not independent.

A collection $A_1, A_2, \dots, A_n$ of events is said to be *pairwise independent* if

$$\Pr\{A_i \cap A_j\} = \Pr\{A_i\} \Pr\{A_j\}$$

for all $1 \leq i < j \leq n$. We say that the events of the collection are **(mutually) independent** if every k -subset $A_{i_1}, A_{i_2}, \dots, A_{i_k}$ of the collection, where $2 \leq k \leq n$ and $1 \leq i_1 < i_2 < \dots < i_k \leq n$, satisfies

$$\Pr\{A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}\} = \Pr\{A_{i_1}\} \Pr\{A_{i_2}\} \dots \Pr\{A_{i_k}\} .$$

For example, suppose we flip two fair coins. Let A_1 be the event that the first coin is heads, let A_2 be the event that the second coin is heads, and let A_3 be the event that the two coins are different. We have

$$\begin{aligned} \Pr\{A_1\} &= 1/2 , \\ \Pr\{A_2\} &= 1/2 , \\ \Pr\{A_3\} &= 1/2 , \\ \Pr\{A_1 \cap A_2\} &= 1/4 , \\ \Pr\{A_1 \cap A_3\} &= 1/4 , \\ \Pr\{A_2 \cap A_3\} &= 1/4 , \\ \Pr\{A_1 \cap A_2 \cap A_3\} &= 0 . \end{aligned}$$

Since for $1 \leq i < j \leq 3$, we have $\Pr\{A_i \cap A_j\} = \Pr\{A_i\} \Pr\{A_j\} = 1/4$, the events A_1, A_2 , and A_3 are pairwise independent. The events are not mutually independent, however, because $\Pr\{A_1 \cap A_2 \cap A_3\} = 0$ and $\Pr\{A_1\} \Pr\{A_2\} \Pr\{A_3\} = 1/8 \neq 0$.

Bayes's theorem

From the definition of conditional probability (C.14) and the commutative law $A \cap B = B \cap A$, it follows that for two events A and B , each with nonzero probability,

$$\begin{aligned}\Pr\{A \cap B\} &= \Pr\{B\} \Pr\{A \mid B\} \\ &= \Pr\{A\} \Pr\{B \mid A\} .\end{aligned}\tag{C.16}$$

Solving for $\Pr\{A \mid B\}$, we obtain

$$\Pr\{A \mid B\} = \frac{\Pr\{A\} \Pr\{B \mid A\}}{\Pr\{B\}} ,\tag{C.17}$$

which is known as **Bayes's theorem**. The denominator $\Pr\{B\}$ is a normalizing constant, which we can reformulate as follows. Since $B = (B \cap A) \cup (B \cap \overline{A})$, and since $B \cap A$ and $B \cap \overline{A}$ are mutually exclusive events,

$$\begin{aligned}\Pr\{B\} &= \Pr\{B \cap A\} + \Pr\{B \cap \overline{A}\} \\ &= \Pr\{A\} \Pr\{B \mid A\} + \Pr\{\overline{A}\} \Pr\{B \mid \overline{A}\} .\end{aligned}$$

Substituting into equation (C.17), we obtain an equivalent form of Bayes's theorem:

$$\Pr\{A \mid B\} = \frac{\Pr\{A\} \Pr\{B \mid A\}}{\Pr\{A\} \Pr\{B \mid A\} + \Pr\{\overline{A}\} \Pr\{B \mid \overline{A}\}} .\tag{C.18}$$

Bayes's theorem can simplify the computing of conditional probabilities. For example, suppose that we have a fair coin and a biased coin that always comes up heads. We run an experiment consisting of three independent events: we choose one of the two coins at random, we flip that coin once, and then we flip it again. Suppose that the coin we have chosen comes up heads both times. What is the probability that it is biased?

We solve this problem using Bayes's theorem. Let A be the event that we choose the biased coin, and let B be the event that the chosen coin comes up heads both times. We wish to determine $\Pr\{A \mid B\}$. We have $\Pr\{A\} = 1/2$, $\Pr\{B \mid A\} = 1$, $\Pr\{\overline{A}\} = 1/2$, and $\Pr\{B \mid \overline{A}\} = 1/4$; hence,

$$\begin{aligned}\Pr\{A \mid B\} &= \frac{(1/2) \cdot 1}{(1/2) \cdot 1 + (1/2) \cdot (1/4)} \\ &= 4/5 .\end{aligned}$$

Exercises

C.2-1

Professor Rosencrantz flips a fair coin once. Professor Guildenstern flips a fair coin twice. What is the probability that Professor Rosencrantz obtains more heads than Professor Guildenstern?

C.2-2

Prove **Boole's inequality**: For any finite or countably infinite sequence of events $A_1, A_2, \dots$,

$$\Pr\{A_1 \cup A_2 \cup \dots\} \leq \Pr\{A_1\} + \Pr\{A_2\} + \dots . \quad (\text{C.19})$$

C.2-3

Suppose we shuffle a deck of 10 cards, each bearing a distinct number from 1 to 10, to mix the cards thoroughly. We then remove three cards, one at a time, from the deck. What is the probability that we select the three cards in sorted (increasing) order?

C.2-4

Prove that

$$\Pr\{A \mid B\} + \Pr\{\bar{A} \mid B\} = 1 .$$

C.2-5

Prove that for any collection of events $A_1, A_2, \dots, A_n$,

$$\Pr\{A_1 \cap A_2 \cap \dots \cap A_n\} = \Pr\{A_1\} \cdot \Pr\{A_2 \mid A_1\} \cdot \Pr\{A_3 \mid A_1 \cap A_2\} \cdots \Pr\{A_n \mid A_1 \cap A_2 \cap \dots \cap A_{n-1}\} .$$

C.2-6 ★

Describe a procedure that takes as input two integers a and b such that $0 < a < b$ and, using fair coin flips, produces as output heads with probability a/b and tails with probability $(b - a)/b$. Give a bound on the expected number of coin flips, which should be $O(1)$. (*Hint*: Represent a/b in binary.)

C.2-7 ★

Show how to construct a set of n events that are pairwise independent but such that no subset of $k > 2$ of them is mutually independent.

C.2-8 ★

Two events A and B are **conditionally independent**, given C , if

$$\Pr\{A \cap B \mid C\} = \Pr\{A \mid C\} \cdot \Pr\{B \mid C\} .$$

Give a simple but nontrivial example of two events that are not independent but are conditionally independent given a third event.

C.2-9 ★

You are a contestant in a game show in which a prize is hidden behind one of three curtains. You will win the prize if you select the correct curtain. After you

have picked one curtain but before the curtain is lifted, the emcee lifts one of the other curtains, knowing that it will reveal an empty stage, and asks if you would like to switch from your current selection to the remaining curtain. How would your chances change if you switch? (This question is the celebrated **Monty Hall problem**, named after a game-show host who often presented contestants with just this dilemma.)

C.2-10 ★

A prison warden has randomly picked one prisoner among three to go free. The other two will be executed. The guard knows which one will go free but is forbidden to give any prisoner information regarding his status. Let us call the prisoners X , Y , and Z . Prisoner X asks the guard privately which of Y or Z will be executed, arguing that since he already knows that at least one of them must die, the guard won't be revealing any information about his own status. The guard tells X that Y is to be executed. Prisoner X feels happier now, since he figures that either he or prisoner Z will go free, which means that his probability of going free is now $1/2$. Is he right, or are his chances still $1/3$? Explain.

C.3 Discrete random variables

A (**discrete**) **random variable** X is a function from a finite or countably infinite sample space S to the real numbers. It associates a real number with each possible outcome of an experiment, which allows us to work with the probability distribution induced on the resulting set of numbers. Random variables can also be defined for uncountably infinite sample spaces, but they raise technical issues that are unnecessary to address for our purposes. Henceforth, we shall assume that random variables are discrete.

For a random variable X and a real number x , we define the event $X = x$ to be $\{s \in S : X(s) = x\}$; thus,

$$\Pr\{X = x\} = \sum_{s \in S: X(s)=x} \Pr\{s\}.$$

The function

$$f(x) = \Pr\{X = x\}$$

is the **probability density function** of the random variable X . From the probability axioms, $\Pr\{X = x\} \geq 0$ and $\sum_x \Pr\{X = x\} = 1$.

As an example, consider the experiment of rolling a pair of ordinary, 6-sided dice. There are 36 possible elementary events in the sample space. We assume

that the probability distribution is uniform, so that each elementary event $s \in S$ is equally likely: $\Pr\{s\} = 1/36$. Define the random variable X to be the *maximum* of the two values showing on the dice. We have $\Pr\{X = 3\} = 5/36$, since X assigns a value of 3 to 5 of the 36 possible elementary events, namely, (1, 3), (2, 3), (3, 3), (3, 2), and (3, 1).

We often define several random variables on the same sample space. If X and Y are random variables, the function

$$f(x, y) = \Pr\{X = x \text{ and } Y = y\}$$

is the **joint probability density function** of X and Y . For a fixed value y ,

$$\Pr\{Y = y\} = \sum_x \Pr\{X = x \text{ and } Y = y\} ,$$

and similarly, for a fixed value x ,

$$\Pr\{X = x\} = \sum_y \Pr\{X = x \text{ and } Y = y\} .$$

Using the definition (C.14) of conditional probability, we have

$$\Pr\{X = x \mid Y = y\} = \frac{\Pr\{X = x \text{ and } Y = y\}}{\Pr\{Y = y\}} .$$

We define two random variables X and Y to be **independent** if for all x and y , the events $X = x$ and $Y = y$ are independent or, equivalently, if for all x and y , we have $\Pr\{X = x \text{ and } Y = y\} = \Pr\{X = x\} \Pr\{Y = y\}$.

Given a set of random variables defined over the same sample space, we can define new random variables as sums, products, or other functions of the original variables.

Expected value of a random variable

The simplest and most useful summary of the distribution of a random variable is the “average” of the values it takes on. The **expected value** (or, synonymously, **expectation** or **mean**) of a discrete random variable X is

$$E[X] = \sum_x x \cdot \Pr\{X = x\} , \tag{C.20}$$

which is well defined if the sum is finite or converges absolutely. Sometimes the expectation of X is denoted by μ_X or, when the random variable is apparent from context, simply by μ .

Consider a game in which you flip two fair coins. You earn \$3 for each head but lose \$2 for each tail. The expected value of the random variable X representing

your earnings is

$$\begin{aligned} E[X] &= 6 \cdot \Pr\{2 \text{ H's}\} + 1 \cdot \Pr\{1 \text{ H, } 1 \text{ T}\} - 4 \cdot \Pr\{2 \text{ T's}\} \\ &= 6(1/4) + 1(1/2) - 4(1/4) \\ &= 1. \end{aligned}$$

The expectation of the sum of two random variables is the sum of their expectations, that is,

$$E[X + Y] = E[X] + E[Y], \quad (\text{C.21})$$

whenever $E[X]$ and $E[Y]$ are defined. We call this property **linearity of expectation**, and it holds even if X and Y are not independent. It also extends to finite and absolutely convergent summations of expectations. Linearity of expectation is the key property that enables us to perform probabilistic analyses by using indicator random variables (see Section 5.2).

If X is any random variable, any function $g(x)$ defines a new random variable $g(X)$. If the expectation of $g(X)$ is defined, then

$$E[g(X)] = \sum_x g(x) \cdot \Pr\{X = x\}.$$

Letting $g(x) = ax$, we have for any constant a ,

$$E[aX] = aE[X]. \quad (\text{C.22})$$

Consequently, expectations are linear: for any two random variables X and Y and any constant a ,

$$E[aX + Y] = aE[X] + E[Y]. \quad (\text{C.23})$$

When two random variables X and Y are independent and each has a defined expectation,

$$\begin{aligned} E[XY] &= \sum_x \sum_y xy \cdot \Pr\{X = x \text{ and } Y = y\} \\ &= \sum_x \sum_y xy \cdot \Pr\{X = x\} \Pr\{Y = y\} \\ &= \left(\sum_x x \cdot \Pr\{X = x\} \right) \left(\sum_y y \cdot \Pr\{Y = y\} \right) \\ &= E[X] E[Y]. \end{aligned}$$

In general, when n random variables $X_1, X_2, \dots, X_n$ are mutually independent,

$$E[X_1 X_2 \cdots X_n] = E[X_1] E[X_2] \cdots E[X_n]. \quad (\text{C.24})$$

When a random variable X takes on values from the set of natural numbers $\mathbb{N} = \{0, 1, 2, \dots\}$, we have a nice formula for its expectation:

$$\begin{aligned}
 E[X] &= \sum_{i=0}^{\infty} i \cdot \Pr\{X = i\} \\
 &= \sum_{i=0}^{\infty} i (\Pr\{X \geq i\} - \Pr\{X \geq i+1\}) \\
 &= \sum_{i=1}^{\infty} \Pr\{X \geq i\} ,
 \end{aligned} \tag{C.25}$$

since each term $\Pr\{X \geq i\}$ is added in i times and subtracted out $i-1$ times (except $\Pr\{X \geq 0\}$, which is added in 0 times and not subtracted out at all).

When we apply a convex function $f(x)$ to a random variable X , **Jensen's inequality** gives us

$$E[f(X)] \geq f(E[X]) , \tag{C.26}$$

provided that the expectations exist and are finite. (A function $f(x)$ is **convex** if for all x and y and for all $0 \leq \lambda \leq 1$, we have $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$.)

Variance and standard deviation

The expected value of a random variable does not tell us how “spread out” the variable’s values are. For example, if we have random variables X and Y for which $\Pr\{X = 1/4\} = \Pr\{X = 3/4\} = 1/2$ and $\Pr\{Y = 0\} = \Pr\{Y = 1\} = 1/2$, then both $E[X]$ and $E[Y]$ are $1/2$, yet the actual values taken on by Y are farther from the mean than the actual values taken on by X .

The notion of variance mathematically expresses how far from the mean a random variable’s values are likely to be. The **variance** of a random variable X with mean $E[X]$ is

$$\begin{aligned}
 \text{Var}[X] &= E[(X - E[X])^2] \\
 &= E[X^2 - 2XE[X] + E^2[X]] \\
 &= E[X^2] - 2E[XE[X]] + E^2[X] \\
 &= E[X^2] - 2E^2[X] + E^2[X] \\
 &= E[X^2] - E^2[X] .
 \end{aligned} \tag{C.27}$$

To justify the equality $E[E^2[X]] = E^2[X]$, note that because $E[X]$ is a real number and not a random variable, so is $E^2[X]$. The equality $E[XE[X]] = E^2[X]$

follows from equation (C.22), with $a = E[X]$. Rewriting equation (C.27) yields an expression for the expectation of the square of a random variable:

$$E[X^2] = \text{Var}[X] + E^2[X] . \quad (\text{C.28})$$

The variance of a random variable X and the variance of aX are related (see Exercise C.3-10):

$$\text{Var}[aX] = a^2 \text{Var}[X] .$$

When X and Y are independent random variables,

$$\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y] .$$

In general, if n random variables $X_1, X_2, \dots, X_n$ are pairwise independent, then

$$\text{Var}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \text{Var}[X_i] . \quad (\text{C.29})$$

The **standard deviation** of a random variable X is the nonnegative square root of the variance of X . The standard deviation of a random variable X is sometimes denoted σ_X or simply σ when the random variable X is understood from context. With this notation, the variance of X is denoted σ^2 .

Exercises

C.3-1

Suppose we roll two ordinary, 6-sided dice. What is the expectation of the sum of the two values showing? What is the expectation of the maximum of the two values showing?

C.3-2

An array $A[1..n]$ contains n distinct numbers that are randomly ordered, with each permutation of the n numbers being equally likely. What is the expectation of the index of the maximum element in the array? What is the expectation of the index of the minimum element in the array?

C.3-3

A carnival game consists of three dice in a cage. A player can bet a dollar on any of the numbers 1 through 6. The cage is shaken, and the payoff is as follows. If the player's number doesn't appear on any of the dice, he loses his dollar. Otherwise, if his number appears on exactly k of the three dice, for $k = 1, 2, 3$, he keeps his dollar and wins k more dollars. What is his expected gain from playing the carnival game once?

C.3-4

Argue that if X and Y are nonnegative random variables, then

$$E[\max(X, Y)] \leq E[X] + E[Y] .$$

C.3-5 ★

Let X and Y be independent random variables. Prove that $f(X)$ and $g(Y)$ are independent for any choice of functions f and g .

C.3-6 ★

Let X be a nonnegative random variable, and suppose that $E[X]$ is well defined. Prove **Markov's inequality**:

$$\Pr\{X \geq t\} \leq E[X] / t \tag{C.30}$$

for all $t > 0$.

C.3-7 ★

Let S be a sample space, and let X and X' be random variables such that $X(s) \geq X'(s)$ for all $s \in S$. Prove that for any real constant t ,

$$\Pr\{X \geq t\} \geq \Pr\{X' \geq t\} .$$

C.3-8

Which is larger: the expectation of the square of a random variable, or the square of its expectation?

C.3-9

Show that for any random variable X that takes on only the values 0 and 1, we have $\text{Var}[X] = E[X]E[1 - X]$.

C.3-10

Prove that $\text{Var}[aX] = a^2\text{Var}[X]$ from the definition (C.27) of variance.

C.4 The geometric and binomial distributions

We can think of a coin flip as an instance of a **Bernoulli trial**, which is an experiment with only two possible outcomes: **success**, which occurs with probability p , and **failure**, which occurs with probability $q = 1 - p$. When we speak of **Bernoulli trials** collectively, we mean that the trials are mutually independent and, unless we specifically say otherwise, that each has the same probability p for success. Two

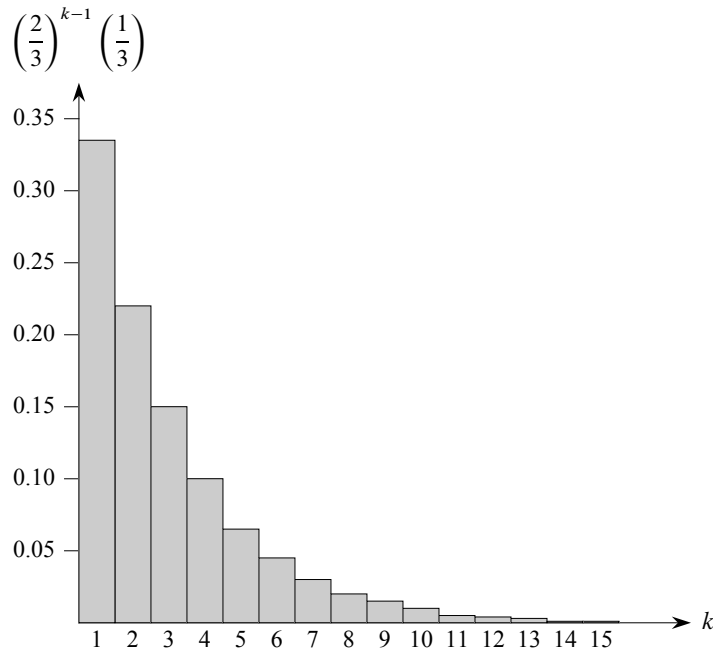


Figure C.1 A geometric distribution with probability $p = 1/3$ of success and a probability $q = 1 - p$ of failure. The expectation of the distribution is $1/p = 3$.

important distributions arise from Bernoulli trials: the geometric distribution and the binomial distribution.

The geometric distribution

Suppose we have a sequence of Bernoulli trials, each with a probability p of success and a probability $q = 1 - p$ of failure. How many trials occur before we obtain a success? Let us define the random variable X be the number of trials needed to obtain a success. Then X has values in the range $\{1, 2, \dots\}$, and for $k \geq 1$,

$$\Pr\{X = k\} = q^{k-1} p, \quad (\text{C.31})$$

since we have $k - 1$ failures before the one success. A probability distribution satisfying equation (C.31) is said to be a **geometric distribution**. Figure C.1 illustrates such a distribution.

Assuming that $q < 1$, we can calculate the expectation of a geometric distribution using identity (A.8):

$$\begin{aligned}
 E[X] &= \sum_{k=1}^{\infty} k q^{k-1} p \\
 &= \frac{p}{q} \sum_{k=0}^{\infty} k q^k \\
 &= \frac{p}{q} \cdot \frac{q}{(1-q)^2} \\
 &= \frac{p}{q} \cdot \frac{q}{p^2} \\
 &= 1/p.
 \end{aligned} \tag{C.32}$$

Thus, on average, it takes $1/p$ trials before we obtain a success, an intuitive result. The variance, which can be calculated similarly, but using Exercise A.1-3, is

$$\text{Var}[X] = q/p^2. \tag{C.33}$$

As an example, suppose we repeatedly roll two dice until we obtain either a seven or an eleven. Of the 36 possible outcomes, 6 yield a seven and 2 yield an eleven. Thus, the probability of success is $p = 8/36 = 2/9$, and we must roll $1/p = 9/2 = 4.5$ times on average to obtain a seven or eleven.

The binomial distribution

How many successes occur during n Bernoulli trials, where a success occurs with probability p and a failure with probability $q = 1 - p$? Define the random variable X to be the number of successes in n trials. Then X has values in the range $\{0, 1, \dots, n\}$, and for $k = 0, 1, \dots, n$,

$$\Pr\{X = k\} = \binom{n}{k} p^k q^{n-k}, \tag{C.34}$$

since there are $\binom{n}{k}$ ways to pick which k of the n trials are successes, and the probability that each occurs is $p^k q^{n-k}$. A probability distribution satisfying equation (C.34) is said to be a **binomial distribution**. For convenience, we define the family of binomial distributions using the notation

$$b(k; n, p) = \binom{n}{k} p^k (1-p)^{n-k}. \tag{C.35}$$

Figure C.2 illustrates a binomial distribution. The name “binomial” comes from the right-hand side of equation (C.34) being the k th term of the expansion of $(p + q)^n$. Consequently, since $p + q = 1$,

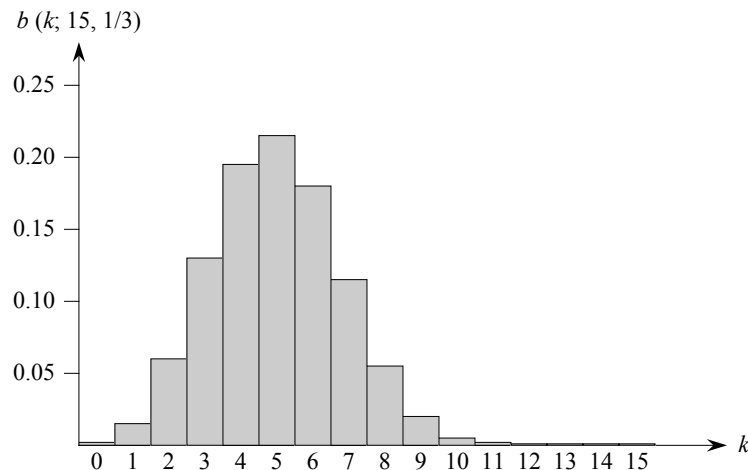


Figure C.2 The binomial distribution $b(k; 15, 1/3)$ resulting from $n = 15$ Bernoulli trials, each with probability $p = 1/3$ of success. The expectation of the distribution is $np = 5$.

$$\sum_{k=0}^n b(k; n, p) = 1, \quad (\text{C.36})$$

as axiom 2 of the probability axioms requires.

We can compute the expectation of a random variable having a binomial distribution from equations (C.8) and (C.36). Let X be a random variable that follows the binomial distribution $b(k; n, p)$, and let $q = 1 - p$. By the definition of expectation, we have

$$\begin{aligned}
 E[X] &= \sum_{k=0}^n k \cdot \Pr\{X = k\} \\
 &= \sum_{k=0}^n k \cdot b(k; n, p) \\
 &= \sum_{k=1}^n k \binom{n}{k} p^k q^{n-k} \\
 &= np \sum_{k=1}^n \binom{n-1}{k-1} p^{k-1} q^{n-k} \quad (\text{by equation (C.8)}) \\
 &= np \sum_{k=0}^{n-1} \binom{n-1}{k} p^k q^{(n-1)-k}
 \end{aligned}$$

$$\begin{aligned}
&= np \sum_{k=0}^{n-1} b(k; n-1, p) \\
&= np \quad (\text{by equation (C.36)}) .
\end{aligned} \tag{C.37}$$

By using the linearity of expectation, we can obtain the same result with substantially less algebra. Let X_i be the random variable describing the number of successes in the i th trial. Then $E[X_i] = p \cdot 1 + q \cdot 0 = p$, and by linearity of expectation (equation (C.21)), the expected number of successes for n trials is

$$\begin{aligned}
E[X] &= E\left[\sum_{i=1}^n X_i\right] \\
&= \sum_{i=1}^n E[X_i] \\
&= \sum_{i=1}^n p \\
&= np .
\end{aligned} \tag{C.38}$$

We can use the same approach to calculate the variance of the distribution. Using equation (C.27), we have $\text{Var}[X_i] = E[X_i^2] - E^2[X_i]$. Since X_i only takes on the values 0 and 1, we have $X_i^2 = X_i$, which implies $E[X_i^2] = E[X_i] = p$. Hence,

$$\text{Var}[X_i] = p - p^2 = p(1 - p) = pq . \tag{C.39}$$

To compute the variance of X , we take advantage of the independence of the n trials; thus, by equation (C.29),

$$\begin{aligned}
\text{Var}[X] &= \text{Var}\left[\sum_{i=1}^n X_i\right] \\
&= \sum_{i=1}^n \text{Var}[X_i] \\
&= \sum_{i=1}^n pq \\
&= npq .
\end{aligned} \tag{C.40}$$

As Figure C.2 shows, the binomial distribution $b(k; n, p)$ increases with k until it reaches the mean np , and then it decreases. We can prove that the distribution always behaves in this manner by looking at the ratio of successive terms:

$$\begin{aligned}
\frac{b(k; n, p)}{b(k-1; n, p)} &= \frac{\binom{n}{k} p^k q^{n-k}}{\binom{n}{k-1} p^{k-1} q^{n-k+1}} \\
&= \frac{n!(k-1)!(n-k+1)!p}{k!(n-k)!n!q} \\
&= \frac{(n-k+1)p}{kq} \\
&= 1 + \frac{(n+1)p-k}{kq}.
\end{aligned} \tag{C.41}$$

This ratio is greater than 1 precisely when $(n+1)p - k$ is positive. Consequently, $b(k; n, p) > b(k-1; n, p)$ for $k < (n+1)p$ (the distribution increases), and $b(k; n, p) < b(k-1; n, p)$ for $k > (n+1)p$ (the distribution decreases). If $k = (n+1)p$ is an integer, then $b(k; n, p) = b(k-1; n, p)$, and so the distribution then has two maxima: at $k = (n+1)p$ and at $k-1 = (n+1)p-1 = np - q$. Otherwise, it attains a maximum at the unique integer k that lies in the range $np - q < k < (n+1)p$.

The following lemma provides an upper bound on the binomial distribution.

Lemma C.1

Let $n \geq 0$, let $0 < p < 1$, let $q = 1 - p$, and let $0 \leq k \leq n$. Then

$$b(k; n, p) \leq \left(\frac{np}{k}\right)^k \left(\frac{nq}{n-k}\right)^{n-k}.$$

Proof Using equation (C.6), we have

$$\begin{aligned}
b(k; n, p) &= \binom{n}{k} p^k q^{n-k} \\
&\leq \left(\frac{n}{k}\right)^k \left(\frac{n}{n-k}\right)^{n-k} p^k q^{n-k} \\
&= \left(\frac{np}{k}\right)^k \left(\frac{nq}{n-k}\right)^{n-k}.
\end{aligned}$$

■

Exercises

C.4-1

Verify axiom 2 of the probability axioms for the geometric distribution.

C.4-2

How many times on average must we flip 6 fair coins before we obtain 3 heads and 3 tails?

C.4-3

Show that $b(k; n, p) = b(n - k; n, q)$, where $q = 1 - p$.

C.4-4

Show that value of the maximum of the binomial distribution $b(k; n, p)$ is approximately $1/\sqrt{2\pi npq}$, where $q = 1 - p$.

C.4-5 ★

Show that the probability of no successes in n Bernoulli trials, each with probability $p = 1/n$, is approximately $1/e$. Show that the probability of exactly one success is also approximately $1/e$.

C.4-6 ★

Professor Rosencrantz flips a fair coin n times, and so does Professor Guildenstern. Show that the probability that they get the same number of heads is $\binom{2n}{n}/4^n$. (*Hint:* For Professor Rosencrantz, call a head a success; for Professor Guildenstern, call a tail a success.) Use your argument to verify the identity

$$\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}.$$

C.4-7 ★

Show that for $0 \leq k \leq n$,

$$b(k; n, 1/2) \leq 2^{n H(k/n) - n},$$

where $H(x)$ is the entropy function (C.7).

C.4-8 ★

Consider n Bernoulli trials, where for $i = 1, 2, \dots, n$, the i th trial has probability p_i of success, and let X be the random variable denoting the total number of successes. Let $p \geq p_i$ for all $i = 1, 2, \dots, n$. Prove that for $1 \leq k \leq n$,

$$\Pr\{X < k\} \geq \sum_{i=0}^{k-1} b(i; n, p).$$

C.4-9 ★

Let X be the random variable for the total number of successes in a set A of n Bernoulli trials, where the i th trial has a probability p_i of success, and let X' be the random variable for the total number of successes in a second set A' of n Bernoulli trials, where the i th trial has a probability $p'_i \geq p_i$ of success. Prove that for $0 \leq k \leq n$,

$$\Pr\{X' \geq k\} \geq \Pr\{X \geq k\}.$$

(Hint: Show how to obtain the Bernoulli trials in A' by an experiment involving the trials of A , and use the result of Exercise C.3-7.)

★ C.5 The tails of the binomial distribution

The probability of having at least, or at most, k successes in n Bernoulli trials, each with probability p of success, is often of more interest than the probability of having exactly k successes. In this section, we investigate the *tails* of the binomial distribution: the two regions of the distribution $b(k; n, p)$ that are far from the mean np . We shall prove several important bounds on (the sum of all terms in) a tail.

We first provide a bound on the right tail of the distribution $b(k; n, p)$. We can determine bounds on the left tail by inverting the roles of successes and failures.

Theorem C.2

Consider a sequence of n Bernoulli trials, where success occurs with probability p . Let X be the random variable denoting the total number of successes. Then for $0 \leq k \leq n$, the probability of at least k successes is

$$\begin{aligned} \Pr\{X \geq k\} &= \sum_{i=k}^n b(i; n, p) \\ &\leq \binom{n}{k} p^k. \end{aligned}$$

Proof For $S \subseteq \{1, 2, \dots, n\}$, we let A_S denote the event that the i th trial is a success for every $i \in S$. Clearly $\Pr\{A_S\} = p^{|S|}$ if $|S| = k$. We have

$$\begin{aligned} \Pr\{X \geq k\} &= \Pr\{\text{there exists } S \subseteq \{1, 2, \dots, n\} : |S| = k \text{ and } A_S\} \\ &= \Pr\left\{\bigcup_{S \subseteq \{1, 2, \dots, n\} : |S|=k} A_S\right\} \\ &\leq \sum_{S \subseteq \{1, 2, \dots, n\} : |S|=k} \Pr\{A_S\} \quad (\text{by inequality (C.19)}) \\ &= \binom{n}{k} p^k. \end{aligned}$$

■

The following corollary restates the theorem for the left tail of the binomial distribution. In general, we shall leave it to you to adapt the proofs from one tail to the other.

Corollary C.3

Consider a sequence of n Bernoulli trials, where success occurs with probability p . If X is the random variable denoting the total number of successes, then for $0 \leq k \leq n$, the probability of at most k successes is

$$\begin{aligned} \Pr\{X \leq k\} &= \sum_{i=0}^k b(i; n, p) \\ &\leq \binom{n}{n-k} (1-p)^{n-k} \\ &= \binom{n}{k} (1-p)^{n-k}. \end{aligned} \quad \blacksquare$$

Our next bound concerns the left tail of the binomial distribution. Its corollary shows that, far from the mean, the left tail diminishes exponentially.

Theorem C.4

Consider a sequence of n Bernoulli trials, where success occurs with probability p and failure with probability $q = 1 - p$. Let X be the random variable denoting the total number of successes. Then for $0 < k < np$, the probability of fewer than k successes is

$$\begin{aligned} \Pr\{X < k\} &= \sum_{i=0}^{k-1} b(i; n, p) \\ &< \frac{kq}{np - k} b(k; n, p). \end{aligned}$$

Proof We bound the series $\sum_{i=0}^{k-1} b(i; n, p)$ by a geometric series using the technique from Section A.2, page 1151. For $i = 1, 2, \dots, k$, we have from equation (C.41),

$$\begin{aligned} \frac{b(i-1; n, p)}{b(i; n, p)} &= \frac{iq}{(n-i+1)p} \\ &< \frac{iq}{(n-i)p} \\ &\leq \frac{kq}{(n-k)p}. \end{aligned}$$